GENERAL:: ENERGY TECHNOLOGIES [EE 403]

1.1 TITLE: Energy Technologies

1.2 *COURSE NUMBER: DE. EE 403.14

1.3 CREDITS : 11 (3-1-0)

1.4 SEMESTER-OFFERED: B.Tech Part-IV (Semester-VII)

1.5 SYLLABUS COMMITTEE MEMBERS : Prof D N Vishwakarma, Dr Manish Kumar

2. OBJECTIVE: Objective of this course is to familiarize students about various conventional and non-conventional resources of energy, their utilization, storage, conservation and management.

3. COURSE CONTENT:

Energy Availability Studies: Forecasting methodologies, technology selection and evaluation, patterns of energy consumptions, supply and Demand projections.

Energy Resources: Concept and classification

Thermal Power Plant: Principle of working, operation of boiler, coal requirement.

Hydro-electric Power Generation: Introduction, principle and working.

Nuclear Power Plant: Principle and working, fission and fusion nuclear reactor, types and description, breeding, nuclear fuel and power control, fuel requirement, thermonuclear fusion.

Non-conventional energy: Introduction to solar, wind, MHD, fuel cells, geothermal, biomass and biogas-design aspects of power plants, applications and environmental aspects.

Ocean energy conversion: Tidal power, wave energy and ocean thermal energy conversion

Direct energy conversion.

Energy storage: Direct and indirect

Concept of energy conservation and management

Economic Consideration.

4. READINGS

4.1 TEXTBOOKS :

1. Non-Conventional Energy Resources by Shobh Nath Singh, Pearson India
2. Renewable Energy Sources and Emerging Technologies by D P Kothari, K C Singal and Rakesh Ranjan, PHI Learning Pvt Ltd.
3. Renewable Energy Resources by John Twidell and Tony Weir, Taylor and Francis

5. OUTCOME OF THE COURSE::